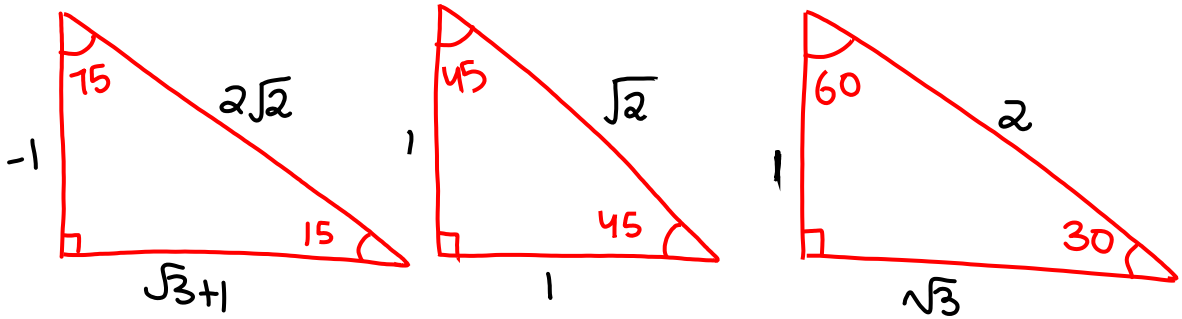
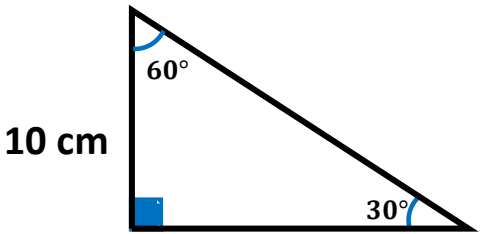
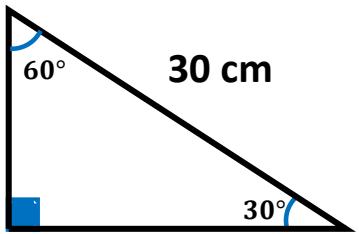
Important or repetitive triangles

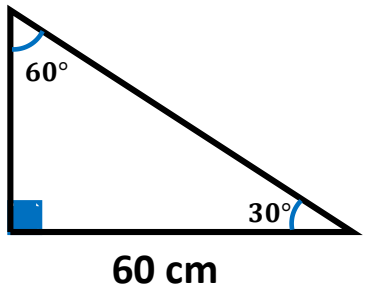
Side ratios of important triangles:

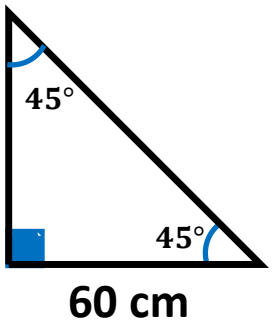


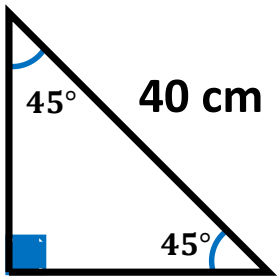
Find the lengths of all the sides of the triangles:

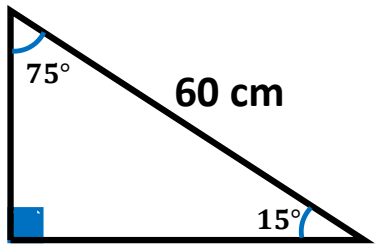




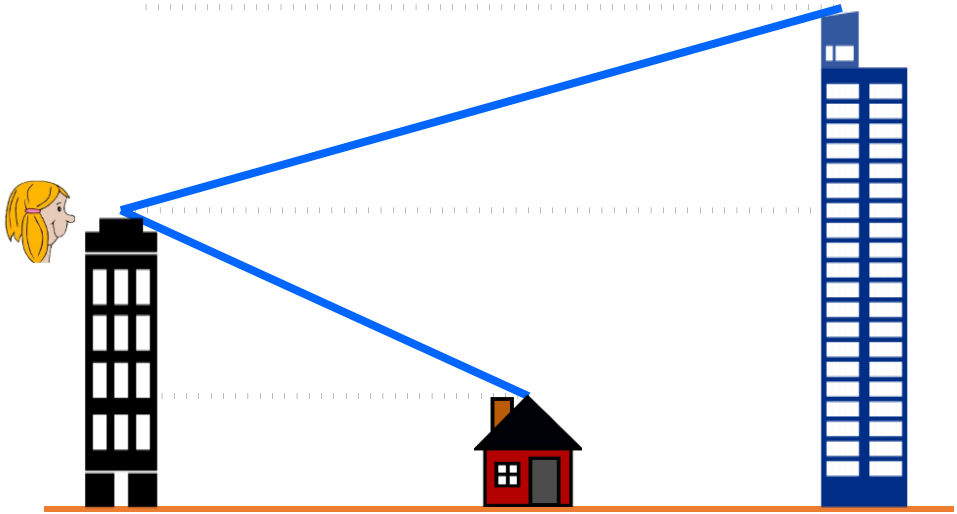


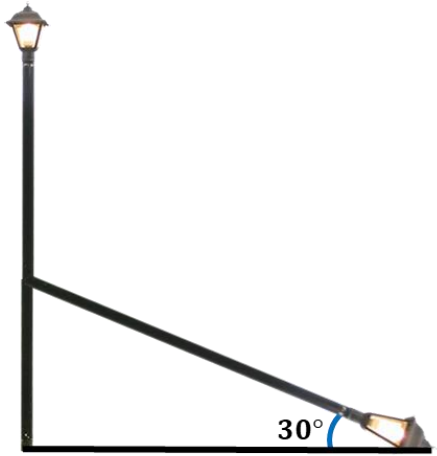






Terminology ( ):



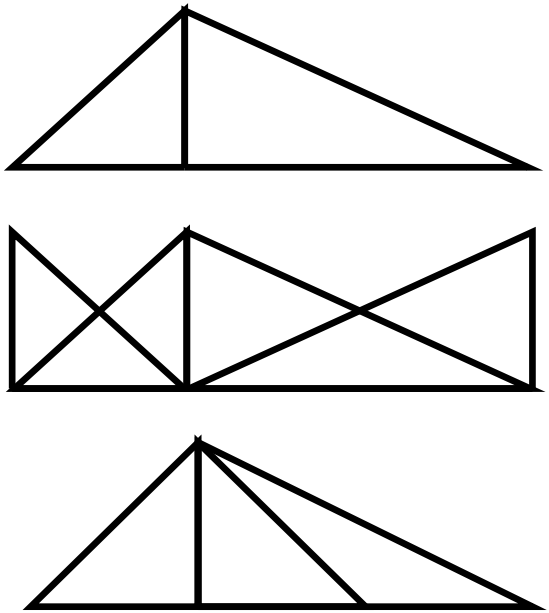


(1) **A vertical post 15 feet high is broken at a certain height and its upper part not completely separated, meets the ground at an angle of 30°. Find the height at which post is broken**

a) 10 b) 5

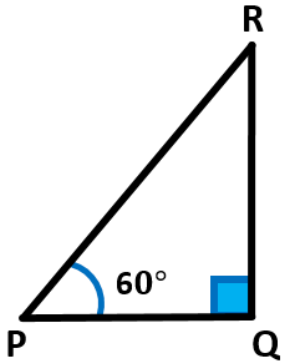
c) $15\sqrt{3}\left(2-\sqrt{3}\right)$ d) $5\sqrt{3}$

Very important short trick:

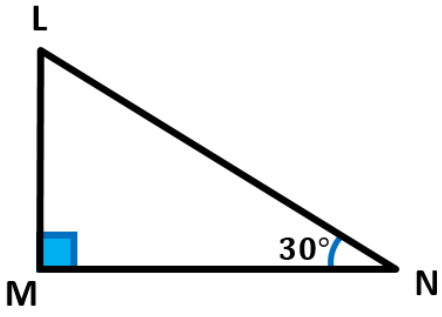


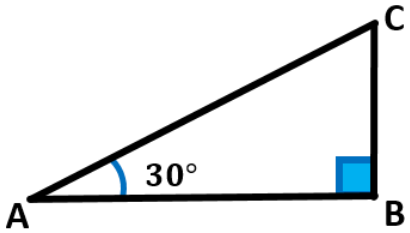
Practice test

1) If $QR = 4\sqrt{3}\ cm$, PR =?



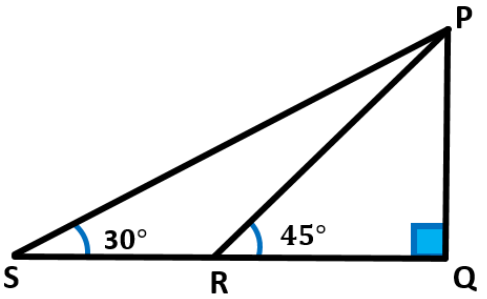
3) If $MN = 30\ cm$, LM =?



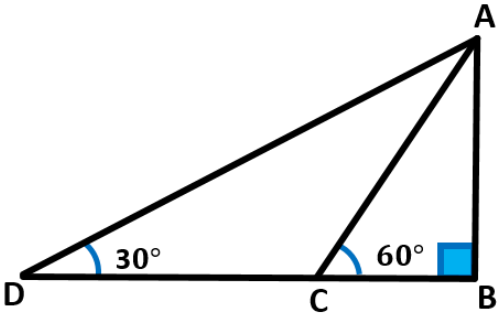


2) If $AB = 15\ cm$, AC =?

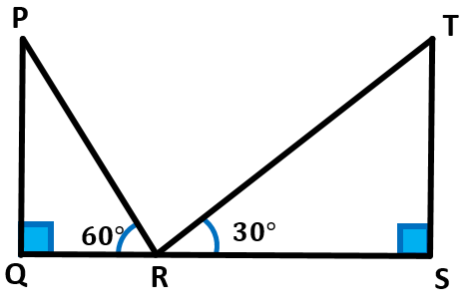
4) If $QR = 20\ cm, SR = ?$



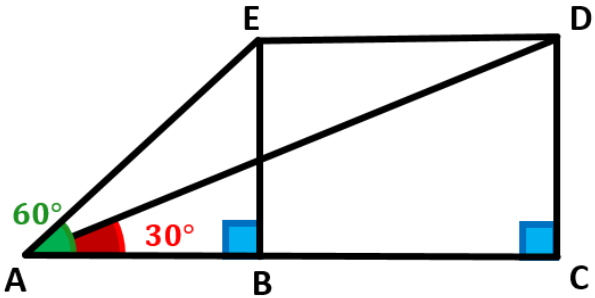
5) If $DC = 15\ cm, AD = ?$



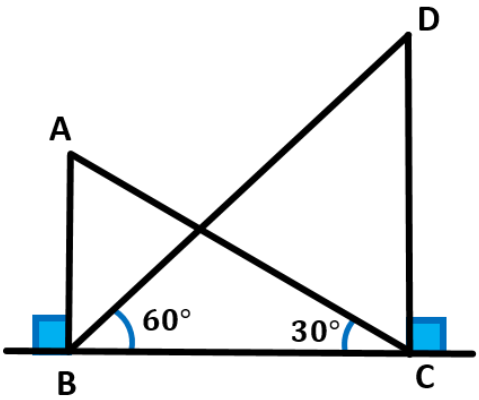
6) If $PQ = TS, QS = 40\ cm, PQ = ?$



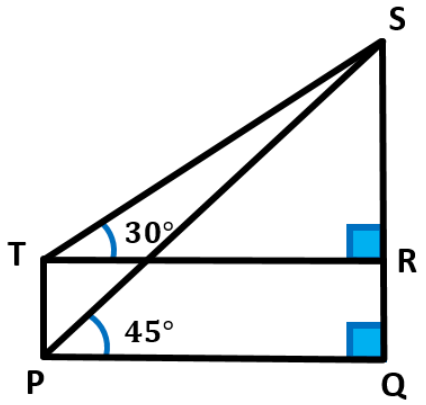
7) If $BE = CD, BC = 30\ cm, BE = ?$



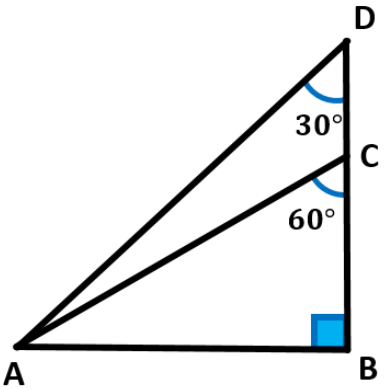
8) $AB : CD = ?$



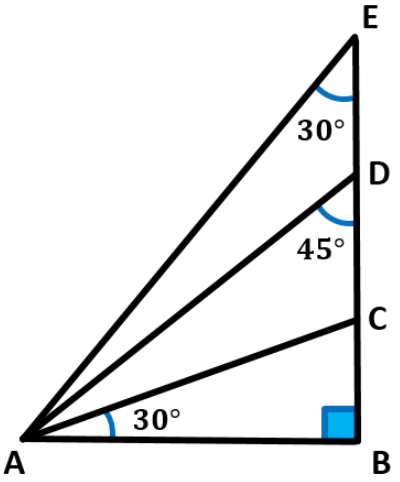
9) If $PT = 100\ cm,\ PT \parallel SQ,\ QS = ?$



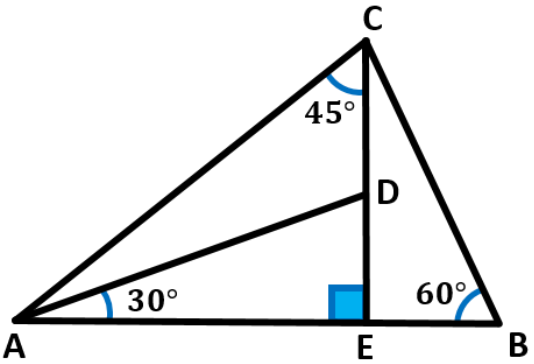
10) If $CD = 60, AD = ?$

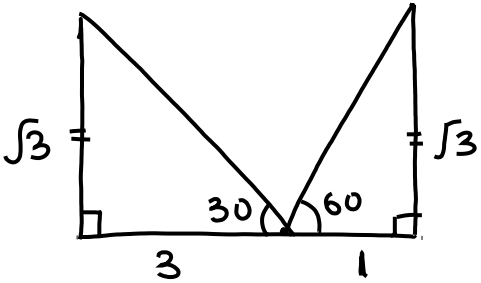


11) $CD : DE = ?$



12) If $CD = 30, AB = ?$

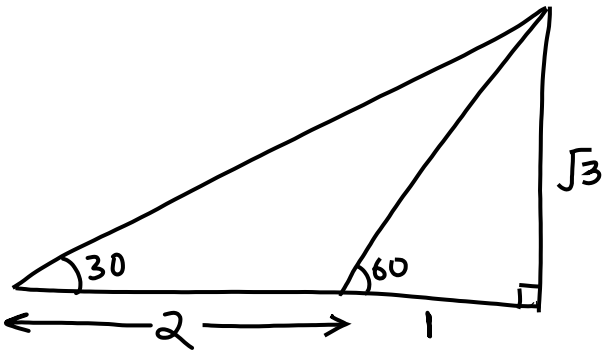




(2) **Two poles of equal height are standing opposite to each other on either side of a road which is 100 m wide. From a point between them on the road, angles of elevation of their tops are 30° and 60°. The height of each pole is**

a) $25\sqrt{3}$ **b)** $20\sqrt{3}$

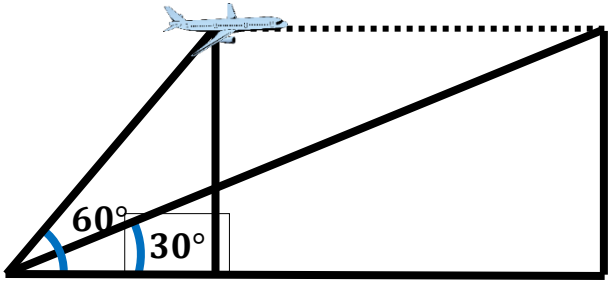
c) 25 **d)** $30\sqrt{3}$



(3) **A man standing at a point P is watching the top of a tower, which makes an angle of elevation of 30°. The man walks some distance towards the tower and then his angle of elevation is 60° . If the height of the tower is 30 m, then the distance moved by him is**

a) 22 **b)** $22\sqrt{3}$

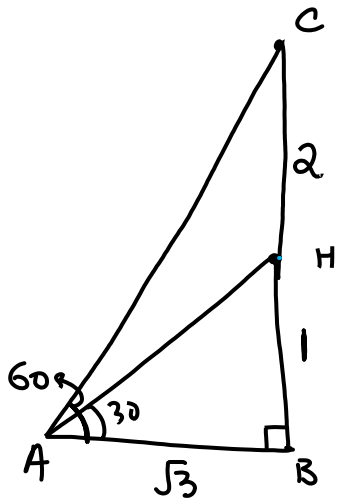
c) 20 **d)** $20\sqrt{3}$



(4) The angle of elevation of an airplane from a point on the ground is 60°. After 15 seconds the elevation changes to 30°. If the airplane is flying at a height of $1500\sqrt{3}$ m. find the speed of the plane.

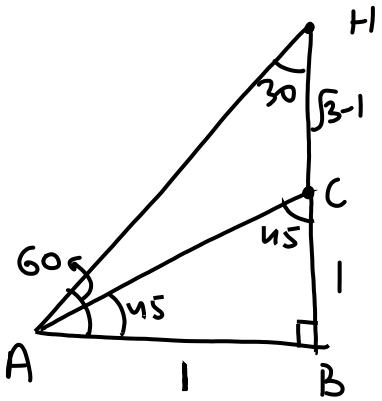
a) 300m/s b) 200m/s
c)100m/s d) 150m/s

Based on SSC CGL 2017 T-2



5. **An airplane when flying at a height of 3125 m from the ground passes vertically below another plane at an instant when the angles of elevation of the two planes from the same point on the ground are 30° and 60° respectively. The distance between two planes at the instant is**

a) 6150 **b) 6000**
c) 5000 **d) 6250**



⑥ **A plane when flying at a height of 6000 m from the ground passes vertically above another plane at an instant, when the angles of elevation of the two planes from the same point on the ground are 60° and 45° respectively. The vertical distance between the planes at that instant is**

a) $3000\left(\sqrt{3}-1\right)$ b) $6000\left(3-\sqrt{3}\right)$

c) $2000\left(3-\sqrt{3}\right)$ d) 4500

7 **If the angle of elevation of the sun changes from 30° to 45 °, the length of shadow of pillar decreases by 20 m. the height of pillar is**

a) $20(\sqrt{3}-1)$

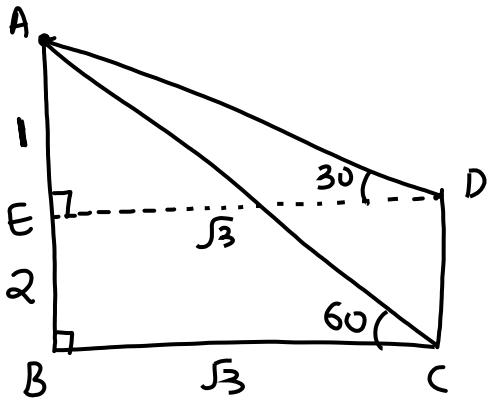
b) $20(\sqrt{3}+1)$

c) $10(\sqrt{3}+1)$

d) $10(\sqrt{3}-1)$

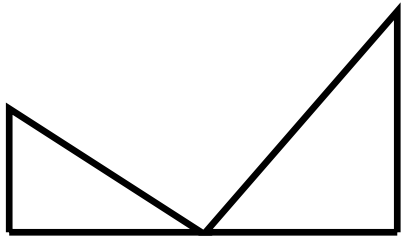
)

)



(8) **There are two vertical post one on each side of a road, just opposite to each other. One post is 108 m high. From the top of this post the angle of depression of the top and foot of the other post are 30° and 60° respectively. The height of other post is**

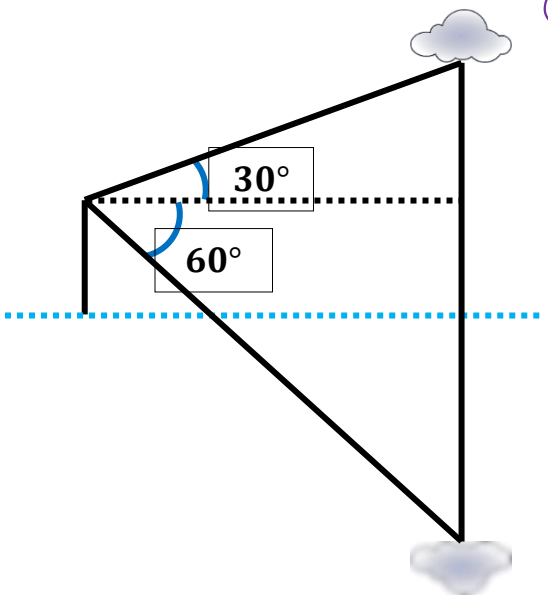
a) 36 **b) 72**
c) 108 **d) 110**



(9) From a point exactly midway between the foot of two towers P and Q, the angles of elevation of their tops are $30°$ and $60°$, respectively. The ratio of the height of P to that of Q is:

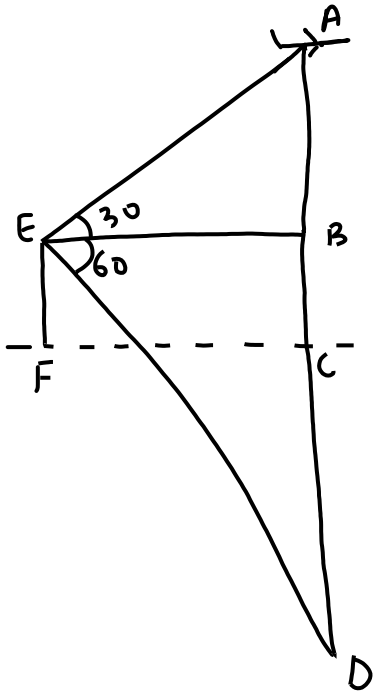
a) $1:3$ b) $1:2$

c) $1:2\sqrt{3}$ d) $2:3\sqrt{3}$



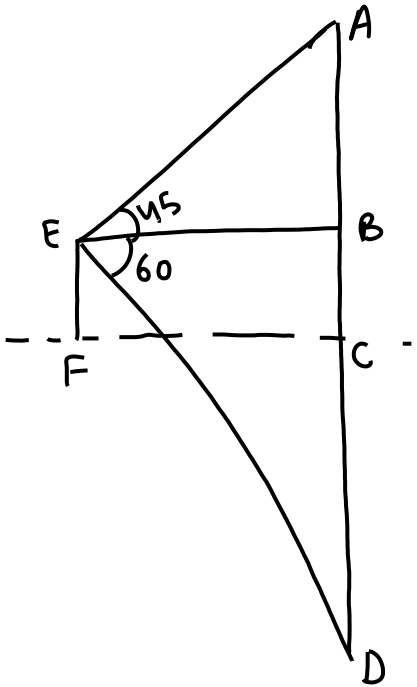
10 The angle of elevation of a cloud from a point 60 m above a lake is $30°$ and the angle of depression of the reflection of the cloud is $60°$. Find the height of the cloud from the surface of the lake.

a) 60 b) 90
c) 120 d) 180



11 HW The angle of elevation of an aeroplane as observed from a point $30\ m$ above the transparent water-surface of a lake is $30°$ and the angle of depression of the image of the aeroplane in the water of the lake is $60°$. The height of the aeroplane from the water-surface of the lake is

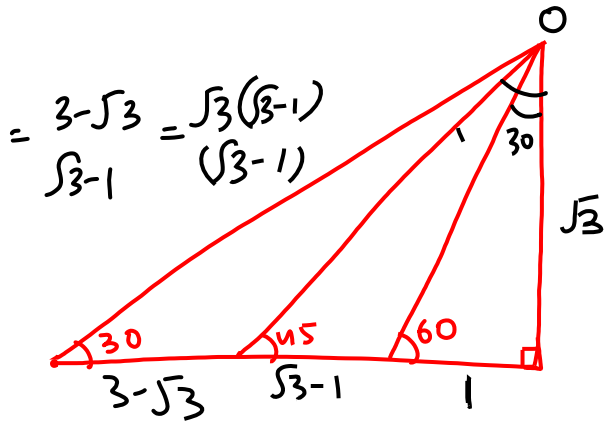
a) $60\ m$ b) $45\ m$
c) $50\ m$ d) $75\ m$



(12) (Hw) **The angle of elevation of a cloud from a point 60 m above a lake is 45° and the angle of depression of the reflection of cloud in the lake is 60°. Find the height of the cloud from the surface of lake?**

a) 223.92 m b) 120 m

c) 204.96 m d) 169.23 m

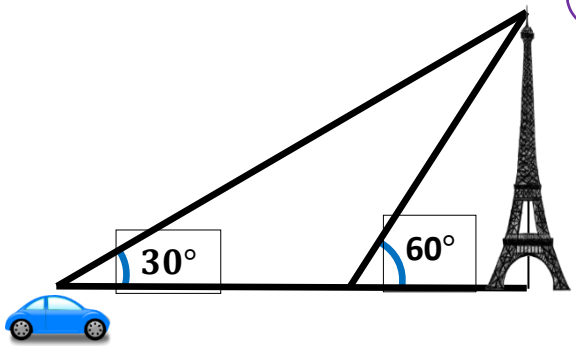


$= \frac{3-\sqrt{3}}{\sqrt{3}-1} = \frac{\sqrt{3}(\sqrt{3}-1)}{(\sqrt{3}-1)}$

(13) If the angles of the elevation of the top of a tower from 3 collinear points A, B and C on a line leading to foot of tower are 30°, 45° and 60° respectively, then find the ratio $AB: BC$?

a) $\sqrt{3}$ b) $2\sqrt{3}-1$

c) $3\sqrt{3}-4$ d) $\frac{4}{\sqrt{3}}$



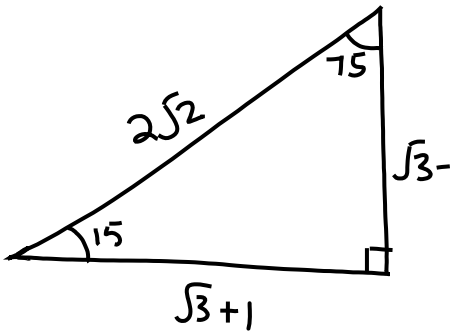
(14) A straight highway leads to the foot of a tower. A man standing at the top of the tower observes a car at an angle of depression of 30°, which is approaching the foot of the tower with a uniform speed. Six seconds later, the angle of depression of the car is found to be 60°. Find the time taken by the car to reach the foot of the tower from this point.

a) 3 sec b) 12 sec
c) 10 sec d) 2 sec

(15) A person from the top of a tower observes a vehicle moving towards him at a uniform speed. It takes 10 minutes for the angle of depression to change from 45° to 60°. After this the time required by the vehicle to reach the bottom of the tower is

a) 12 minutes 20 seconds
b) 13 minutes
c) 13 minutes 40 seconds
d) 14 minutes 24 seconds

SSC CGL 2016 T-2



16 **A ladder is placed against a wall such that it just reaches the top of the wall. The foot of the ladder is at 5 meters from the wall. the angle of elevation of the top of the wall from the base of the ladder is $15°$. What is he length of the ladder?**

a) $5\sqrt{6} - 5\sqrt{3}$ b) $5\sqrt{6} - 5\sqrt{2}$

c) $5\sqrt{2} - 1$ d) $5\sqrt{3} + 5\sqrt{2}$

SSC CGL 2017 T-2